

# ALAN ORDORICA

Software Engineer | Backend Systems | Cloud & Platform

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## Summary

**Multipotentialite**<sup>1</sup> software engineer with a passion for learning and building across disciplines—quickly picking up tools from backend systems to embedded hardware and turning ambiguous challenges into deliverable solutions

## Experience

Google

Sunnyvale, CA

Software Engineer | Google Cloud

May 2022 - Dec 2024

- Designed and implemented a high-fidelity control-plane simulator in C++/Python using gRPC and Protobuf, complete with pub/sub messaging and concurrency primitives (channels, callbacks). This simulator replicated complex Andromeda network behaviors, enabling developers to test new features in isolation and reducing full regression runtimes by **25%** (saving 1.5 SWE-years each year).
- Developed a multicast VPC-gateway emulation layer by extending the simulator to support 1,000+ gateways connected to a single controller. Created new C++ modules and Python orchestration scripts to emit and process messages, mocked missing data-plane components (GSLB, ESF port), and helped customers validate scalable pub/sub topologies—reducing customer onboarding time by **30%**.
- Replaced brittle monitoring hooks with modern observability: migrated from BNS watchers to Streamz/PK-based monitors. Wrote Python scripts to query Monarch for job health, parse Borg metadata, and auto-restart failed services; integrated these into the test harness, cutting device downtime by **27%** and eliminating false alarms.
- Built debugging and orchestration tools for isolated test sandboxes. Provisioned per-test sandboxes in Borg/Sandman, scripted metrics collection and auto-recovery, and authored log-parsing utilities that quickly pinpointed broken runfile paths or output misconfigurations—reducing release-blocking test failures by **30%** and speeding developer feedback loops by **40%**.
- Created pre-population strategies for Spanner test databases. Developed Python utilities to seed test DBs via textproto, Spanner backups, and Sandman snapshots; configured cron jobs and multi-sandbox topologies to ensure reproducible datasets and cut environment setup time significantly.
- Contributed to host-agent re-architecture, replacing an OpenFlow-based on-host agent with a new C++ On-Host Controller (OHC), integrating control-plane logic directly on VM hosts. Updated gRPC interfaces and host-path plumbing, improving reliability and deprecating legacy code.
- Authored design docs and diagrams for VPC-gateway and ESF-port integrations, weighing separate vs. co-located components and collaborating with cross-functional teams to finalize scalable architectures.

Apple

Sunnyvale, CA

Tools and Automation Engineer | CI Tools-Site Reliability

July 2020 - March 2021

- Deployed 200 test racks (1,000+ devices) and documented host configuration strategies tailored to rack use cases; standardized bring-up and handoff processes that reduced scheduling friction for streamlined lab operations by new techs without escalations.
- Automated provisioning (Python/Bash/Ansible) for firmware flashing and network bootstrap; delivered a repeatable playbook and batch orchestration so racks could be brought online consistently without manual intervention.
- Built Splunk dashboards, alerts, and auto-remediation scripts for device health and job status; reduced MTTD/MTTR by enabling on-call techs to triage from a single pane of glass with documented runbooks.
- Defined rack bring-up smoke tests (power, network, device enumeration) and acceptance criteria with SRE/lab ops; ensured racks met production-ready standards before handoff.
- Authored setup docs and remediation runbooks and trained lab techs; improved shift-to-shift continuity and reduced escalations for common failures.

Full Stack Developer

Remote

Software Engineer

January 2020 - Present

- Developed modular React frontend and Node.js backend portfolio web apps with Docker Compose for local orchestration, enabling rapid environment setup.
- Developed Node.js RESTful microservices and ESP32-based automation projects using Python, Bash, and other tools, showcasing proficiency in hardware-software integration and system-level programming.

## Technical Skills

<sup>1</sup> **Multipotentialite**: a person with multiple interests and an ability to learn quickly across different domains

**Languages:** C/C++, Python, JavaScript, Dart, Java, SQL, Bash, JSON, YAML

**Cloud & Distributed Systems:** GCP, Kubernetes, Docker, gRPC, Protobuf, CI/CD, Microservices, Distributed Systems

**DevOps & Build:** Ansible, CMake, Git

**Libraries & Frameworks:** Flutter, React, Next.js, Express.js, ESP-IDF, FreeRTOS

**Embedded Systems:** ESP32, I<sup>2</sup>C, SPI, BLE, GDB

**Education**

**San Francisco State University** San Francisco, CA

Bachelor of Science Computer Engineering